METRIC

DOD-C-63976 (AR)
AMENDMENT 5
29 November 1994
SUPERSEDING
AMENDMENT 4
30 March 1989

MILITARY SPECIFICATION

CASE, CARTRIDGE, 30MM AMMUNITION
FOR M788, M789 AND M799
SERIES OF 30MM AMMUNITION

This amendment forms a part of DOD-C-63976 (AR), dated 14 May 1982, and is approved for use by the US Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.1: Add "MIL-STD-1235 - Single and Multi-Level Continuous Sampling Procedures and Tables for Inspection by Attributes" to the end of the paragraph.

Delete: "FED-STD-115" and substitute "FED-STD-151". 2.1.2: Under "Drawings":

Delete "11825969 - Test Barrel, 30mm Mann (6.5)" and substitute "9390627 - Barrel, Test 30mm (ADEN. Barrel)"

Add dwg: "9390748 - Barrel, Test: Gun, M230, 30mm (Progressive w/6.5 degree exit twist)"

3.2.1: Delete in its entirety and substitute the following:

"3.2.1 Performance. The cartridge case shall meet the following performance requirements when assembled in a 30mm target practice cartridge as shown on Drawing 11825947. Unless otherwise specified, performance requirement involving the firing of the cartridge apply when fired in a test barrel conforming to Drawing 9390748."

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3.2.1.1: Delete in its entirety and substitute the following:

"3.2.1.1 Case rupture When fired under the conditions of 3.2.1, the case shall not rupture. It shall also not rupture when assembled in specially loaded cartridges and subjected to an over-pressure condition from 395 to 430 MPa."

PAGE 5

Note 1/: Delete "4.5.3.10.2" and substitute "4.5.3.10.1".

PAGE 8

3.2.3.1.3: Delete "MIL-STD-331, Method 107" and substitute "MILSTD-810, Method 509".

PAGE 12

Table III: Delete sample Size "20" and substitute "Sample size "60" for case extraction".

PAGE 13

4.4.2a: Add at the end of the sentence "or MIL-STD-1235".

PAGE 14

4.4.2.1: Delete "3/" from Major 104 and add "3/" to Majors l0l, 102, 103.

PAGE 15

4.4.2.1, Minor 222: Delete “3.3.3” and substitute "3.5"

PAGE 16

Note 3/: Delete "min. sample size 315 no defects allowed" and substitute "Sample size determined by .25 AQL (315 min sample size) zero defects allowed."

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* 4.5.1.1: Delete in its entirety and substitute the following:

"4.5.1.1 Case rupture. The round shall be fired in a test barrel which conforms to drawing 9390748. The case shall be visually examined for evidence of case rupture.

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4.5.1.1.1: Delete in its entirety and substitute the following:

"4.5.1.1.1 Chamber over-pressure Special test rounds shall be assembled in which the propellant load is such that the chamber pressure, measured at the case mouth, is within the range of 395 to 430 MPa. The round shall be fired in a test barrel (Dwg 9390748). The cartridge cases shall be examined for evidence of case rupture. The case shall be considered to have failed if it ruptures at a pressure of less than 430 MPa. Cases tested at pressure in excess of 395 MPa shall be considered a success if no failure occurs and will be considered a no-test if failure occurs at pressures above 430 MPa."

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4.5.2.3: Delete "MIL-STD-331 test method 107" and substitute "MIL-STD-810 test method 509 (Slight Mouth Erosion Permissible)

Delete "4.3.2" and substitute "4.5.3"

PAGE 22

4.5.3.4.1, third sentence: Delete "hole" and substitute "hold".

Delete last sentence of last paragraph and substitute "All mechanical properties test results shall be reported."

PAGE 23

4.5.3.4.2, third sentence: Delete "avoided" and substitute avoided".

Delete last sentence and substitute "All hardness test results shall be reported".

PAGE 24

4.5.3.5: Delete last sentence and substitute "The thickness values obtained by both methods should be reported."

4.5.3.4.2: Delete last sentence and substitute "The coating weight data obtained will be reported."

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4.5.3.6: Delete "3.5.3" and substitute "3.5".

4.5.3.7.1: Delete in its entirety and substitute the following.

"4.5.3.7.1: External defects, case sidewall. Examination of the case sidewall for longitudinal cracks or seams shall be performed using the five diopter magnifier or a (NDT) device capable of detecting external cracks or seams. Prior to the examination the entire surface of each case will be examined for random gross defects. When using the optical method, adequate lighting must be available and higher magnification can be used to differentiate between a scratch or a crack. When it is still questionable whether or not a crack exists, then die penetrant testing per MIL-16866 can be employed. Any case found to have a crack in excess of 38.l mm long by .19mm deep shall be considered as a defect. The cross sectioning technique may be used to verify the depth of a crack identified visually.

When a (NDT) method is used, sufficient standards will be available to verify that cracks or seams in excess of .19mm deep by 6mm long can be reliably identified. Any cracks or seams in excess of 19mm deep by 6mm long shall be considered as defects. The area of inspection to be included in the NDT technique shall extend from 15.25mm to 85.OOmm when measured from the head end of the case."

PAGE 26

4.5.3.7.2: Delete "Binocular microscope using a minimum of 7x magnification" and substitute "five diopter magnifier".

Delete "Refer to 6.7" from the end of paragraph 1 and add "Refer to 6.7" to the end of the second paragraph.

4.5.3.7.3, last sentence, second paragraph: Delete "For definition, refer to 6.3.7".

4.5.3.8: Delete last sentence and substitute "The individual values thus obtained shall be reported."

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4.5.3.10: Delete "6.3.8" and substitute "6.8".

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* 6.3: Add sentence to the end of paragraph as follows:

"The letter of submittal shall state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph."

Delete "DRDAR-QAF-I" and substitute "AMSTA-AR-QAC-C". 6.5: Delete in its entirety and substitute the following:

"6.5 Data report. All data obtained from testing as described in 4.4.3.4 and 4.4.3.6 shall be reported."

6.7: Delete first sentence and substitute "The contractor and supplier shall establish defect standards for use in the NDT test defined in 4.3.3.7.1, 4.3.3.7.2 and 4.3.3.7.3."

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6.10: Delete "4.3.4" and substitute "4.5.1 and 4.5.2"

6.11: Delete in its entirety

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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